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(54) **GOLF CLUB HEADS WITH OPTIMIZED CHARACTERISTICS AND RELATED METHODS**

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See application file for complete search history.

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(57)

ABSTRACT

Embodiments of golf club heads comprising a head interior and a hosel structure, the head interior bounded by a head front portion, a head rear portion, a head heel portion, a head toe portion, a head top portion, and a head sole portion, the golf club heads further comprising various optimized characteristics, are presented herein. Other examples and related methods are also disclosed herein.

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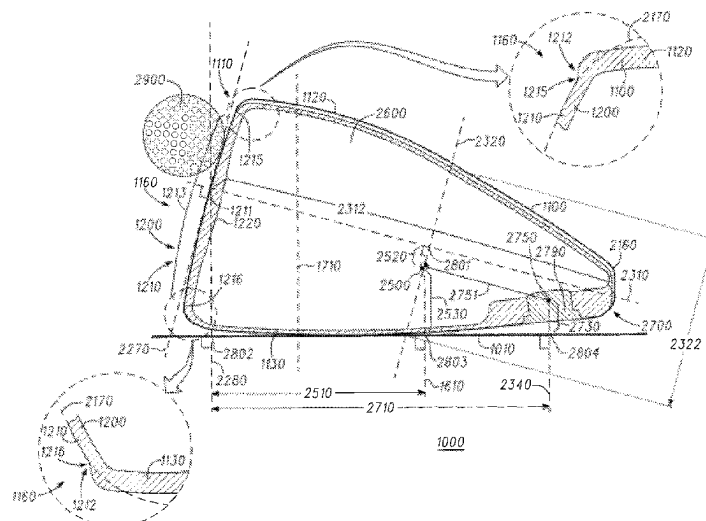
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continuation of application No. 15/680,414, filed on Aug. 18, 2017, now Pat. No. 10,357,700, which is a continuation of application No. 14/836,732, filed on Aug. 26, 2015, now Pat. No. 9,764,205, which is a continuation of application No. 13/804,917, filed on Mar. 14, 2013, now Pat. No. 9,168,429.

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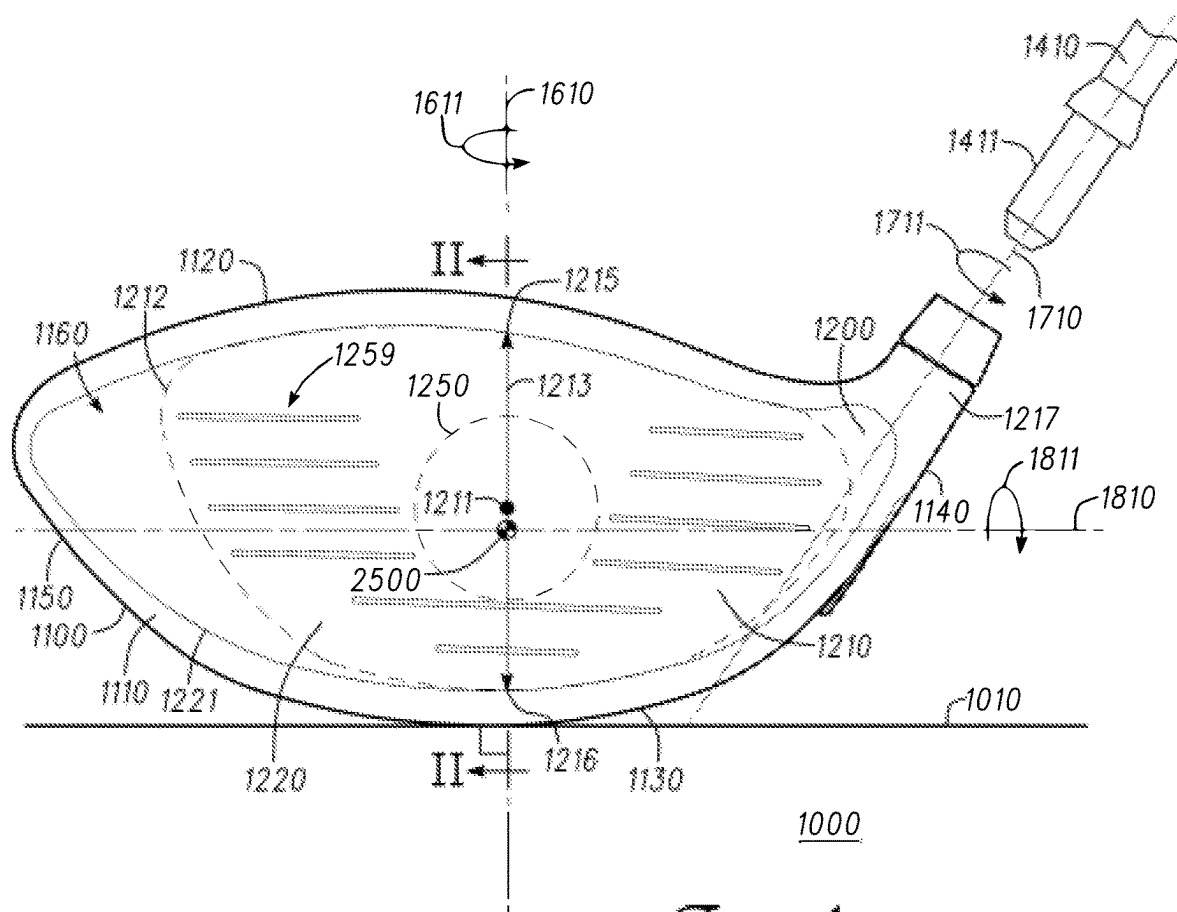
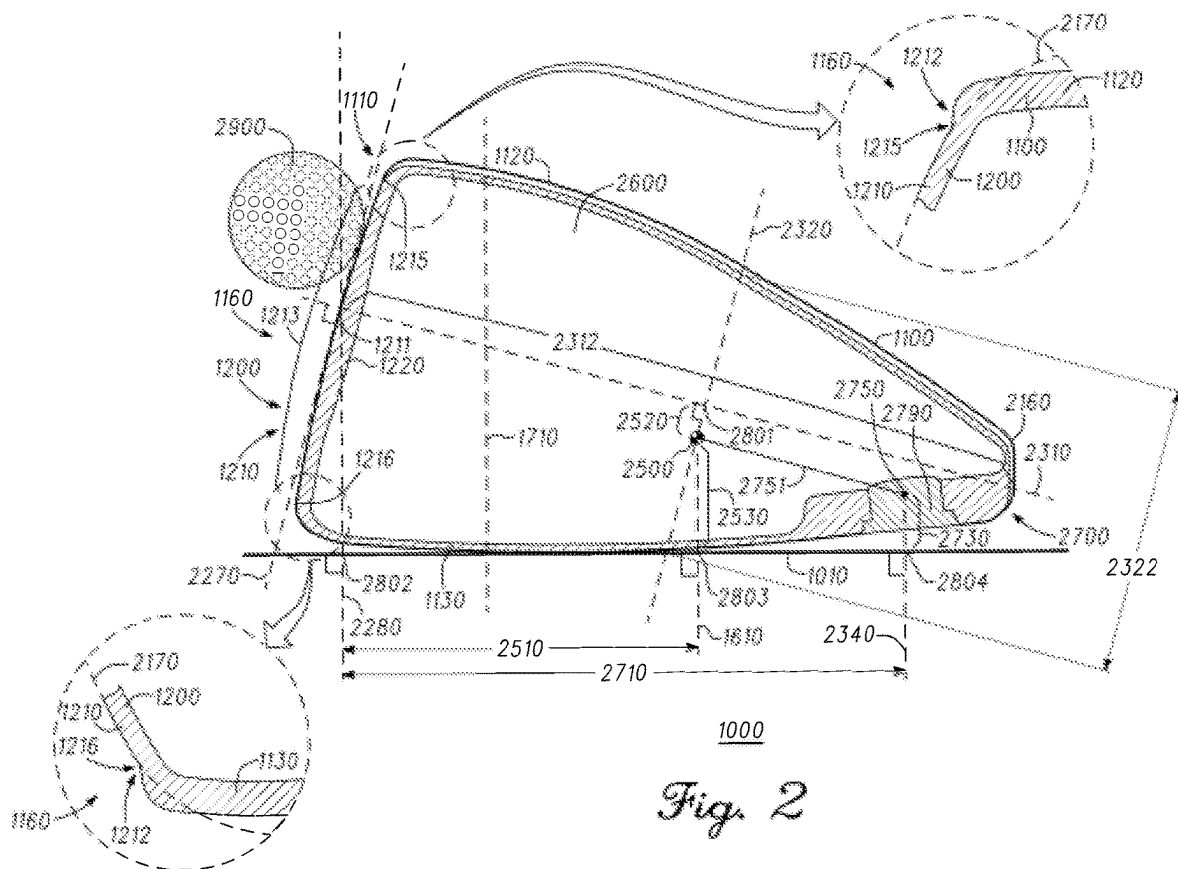


Fig. 1



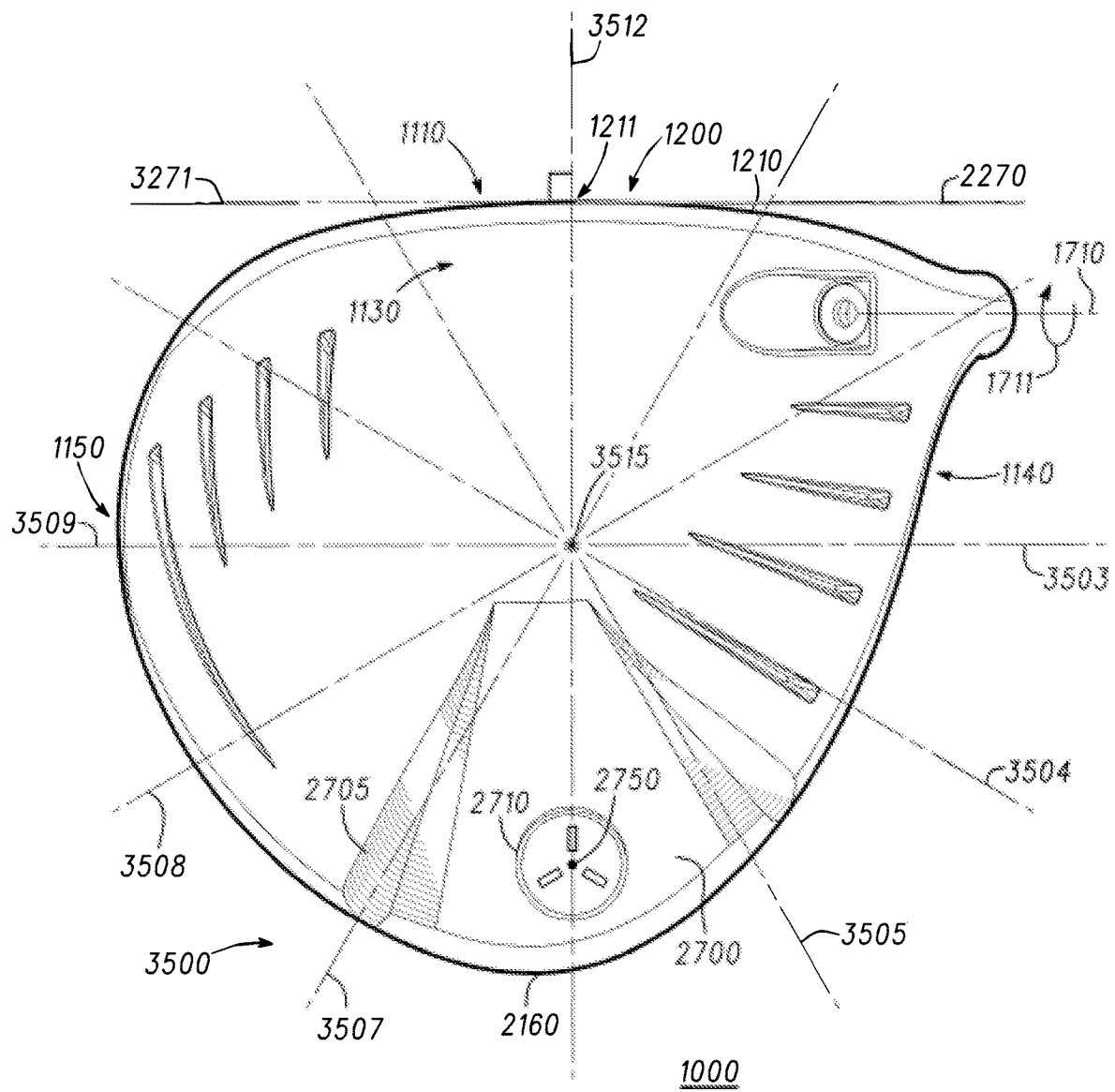
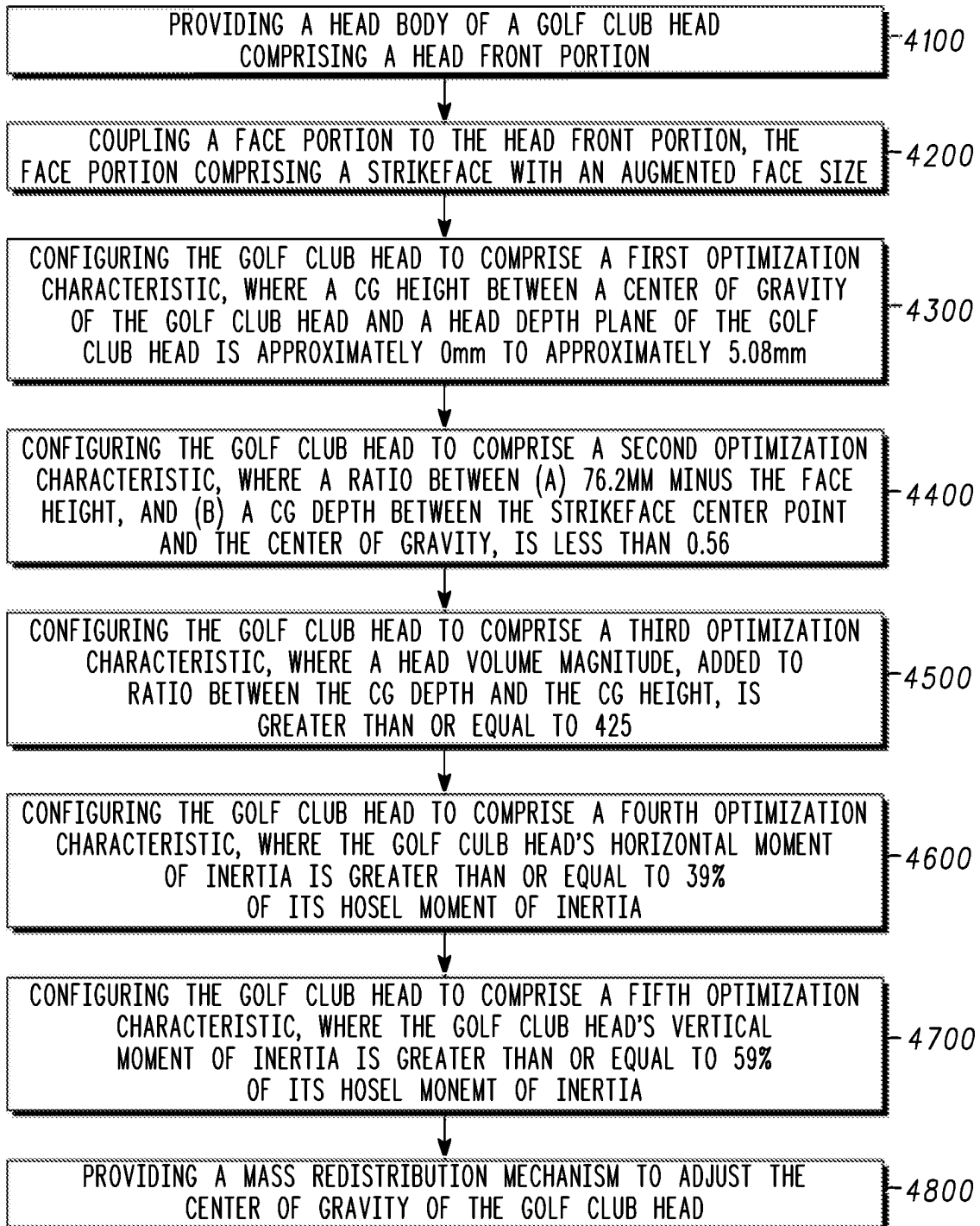


Fig. 3

4000*Fig. 4*

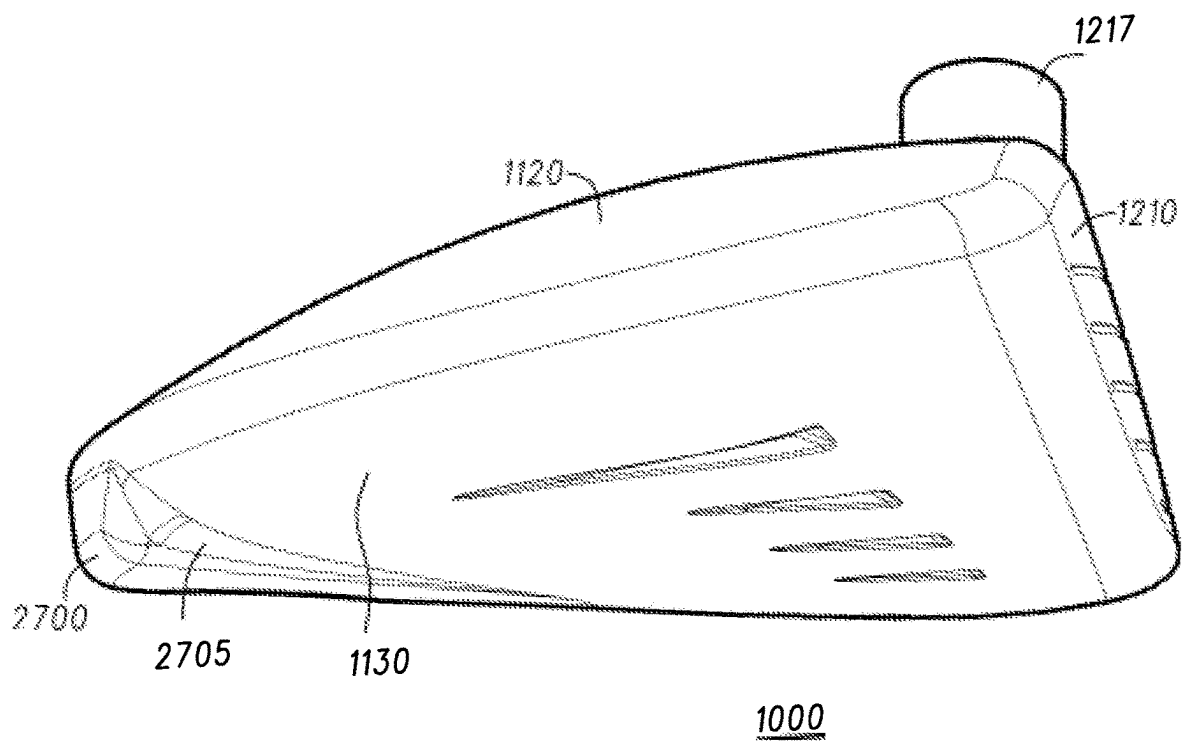


Fig. 5

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GOLF CLUB HEADS WITH OPTIMIZED CHARACTERISTICS AND RELATED METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

This is a continuation of U.S. patent application Ser. No. 17/661,903, filed May 3, 2022, which is a continuation of U.S. patent application Ser. No. 17/094,707, filed on Nov. 10, 2020, now U.S. Pat. No. 11,318,359, issued on May 3, 2022, which is a continuation of U.S. patent application Ser. No. 16/503,322, filed Jul. 3, 2019, now U.S. Pat. No. 10,857,435, issued on Dec. 8, 2020, which is a continuation of U.S. patent application Ser. No. 15/680,414, filed on Aug. 18, 2017, now U.S. Pat. No. 10,357,700, issued on Jul. 23, 2019, which is a continuation of U.S. patent application Ser. No. 14/836,732, filed on Aug. 26, 2015, now U.S. Pat. No. 9,764,205, issued Sep. 19, 2017, which is a continuation of U.S. patent application Ser. No. 13/804,917, filed on Mar. 14, 2013, now U.S. Pat. No. 9,168,429, issued on Oct. 27, 2015, the contents of which are incorporated fully herein by reference.

TECHNICAL FIELD

The present disclosure relates generally to sports equipment, and relates, more particularly, to golf club heads with optimized characteristics and related methods.

BACKGROUND

Golf club heads often comprise different features that can be designed or configured to improve one or more of their performance characteristics. Innate interplay between such different features often exists, however, such that adjusting or configuring one feature may inherently alter another feature, often disadvantageously. As an example, expanding the strikeface of a golf club to provide a greater impact area can alter the location of the center of gravity of the golf club disadvantageously, and unintended performance consequences may ensue if features are not configured or designed in a balanced manner to account for the interplay between the different features.

Considering the above, further developments with respect to golf club features that are balanced with respect to each other will enhance the performance of golf clubs.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure may be better understood from a reading of the following detailed description of examples of embodiments, taken in conjunction with the accompanying figures.

FIG. 1 illustrates a front view of a golf club head in accordance with the present disclosure.

FIG. 2 illustrates a side cross-sectional view of the golf club head along line II-II of FIG. 1.

FIG. 3 illustrates a bottom view of the golf club head of FIGS. 1-2.

FIG. 4 illustrates a flowchart for a method that can be used to provide, form, and/or manufacture a golf club head in accordance with the present disclosure.

FIG. 5 illustrates a side view of the golf club head of FIGS. 1-3.

For simplicity and clarity of illustration, the drawing figures illustrate the general manner of construction, and

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descriptions and details of well-known features and techniques may be omitted to avoid unnecessarily obscuring the present disclosure. Additionally, elements in the drawing figures are not necessarily drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help improve understanding of embodiments of the present disclosure. The same reference numerals in different figures denote the same elements.

The terms “first,” “second,” “third,” “fourth,” and the like in the description and in the claims, if any, are used for distinguishing between similar elements and not necessarily for describing a particular sequential or chronological order. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments described herein are, for example, capable of operation in sequences other than those illustrated or otherwise described herein. Furthermore, the terms “include,” and “have,” and any variations thereof, are intended to cover a non-exclusive inclusion, such that a process, method, system, article, device, or apparatus that comprises a list of elements is not necessarily limited to those elements, but may include other elements not expressly listed or inherent to such process, method, system, article, device, or apparatus.

The terms “left,” “right,” “front,” “back,” “top,” “bottom,” “over,” “under,” and the like in the description and in the claims, if any, are used for descriptive purposes and not necessarily for describing permanent relative positions. It is to be understood that the terms so used are interchangeable under appropriate circumstances such that the embodiments of the apparatus, methods, and/or articles of manufacture described herein are, for example, capable of operation in other orientations than those illustrated or otherwise described herein.

The terms “couple,” “coupled,” “couples,” “coupling,” and the like should be broadly understood and refer to connecting two or more elements, mechanically or otherwise. Coupling (whether mechanical or otherwise) may be for any length of time, e.g., permanent or semi-permanent or only for an instant.

The absence of the word “removably,” “removable,” and the like near the word “coupled,” and the like does not mean that the coupling, etc. in question is or is not removable.

As defined herein, two or more elements are “integral” if they are comprised of the same piece of material. As defined herein, two or more elements are “non-integral” if each is comprised of a different piece of material.

DESCRIPTION

In one example, a golf club head can comprise a head body comprising a head interior and a hosel structure. The head interior can be bounded by a head front portion, a head rear portion, a head heel portion, a head toe portion, a head top portion, and a head sole portion. The hosel structure can have a bore for receiving a golf club shaft, where the bore can have a hosel axis. The golf club head can also comprise a head center of gravity, a head horizontal axis extending through the head center of gravity, from the head heel portion to the head toe portion, and parallel to a ground plane when the golf club head is at an address position over the ground plane, a hosel moment of inertia about the hosel axis, and a horizontal moment of inertia about the head horizontal axis. The horizontal moment of inertia can be greater than or equal to 39% of the hosel moment of inertia.

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In one example, a golf club head can comprise a head body comprising a head interior and a hosel structure. The head interior can be bounded by a head front portion, a head rear portion, a head heel portion, a head toe portion, a head top portion, and a head sole portion. The hosel structure can have a bore for receiving a golf club shaft, where the bore can have a hosel axis. The golf club head can also comprise a head center of gravity, a head vertical axis extending through the head center of gravity, from the head top portion to the head sole portion, and perpendicular to a ground plane when the golf club head is at address over the ground plane, a hosel moment of inertia about the hosel axis, and a vertical moment of inertia about the head vertical axis. The vertical moment of inertia can be greater than or equal to 59% of the hosel moment of inertia.

In one implementation, a method for providing a golf club head can comprise providing a head body having a head interior and a hosel structure. The head interior can be bounded by a head front portion, a head rear portion, a head heel portion, a head toe portion, a head top portion, and a head sole portion. The hosel structure can have a bore for receiving a golf club shaft, where the bore can have a hosel axis. The method can also comprise coupling the golf club shaft to the hosel structure. A head horizontal axis can extend through a head center of gravity of the golf club head, from the head heel portion to the head toe portion, and parallel to a ground plane when the golf club head is at an address position over the ground plane. A head vertical axis can extend through the head center of gravity, from the head top portion to the head sole portion, and perpendicular to the ground plane when the golf club head is at address over the ground plane. In addition, providing the head body can comprise at least one of: (a) establishing a horizontal moment of inertia about the head horizontal axis to be greater than or equal to 39% of a hosel moment of inertia about the hosel axis, or (b) establishing a vertical moment of inertia about the head vertical axis to be greater than or equal to 59% of the hosel moment of inertia about the hosel axis.

In one example, a golf club head can comprise a head body comprising a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, where the bore can have a hosel axis. The golf club head can also comprise a strikeface at the head front portion and comprising a strikeface centerpoint, a head volume measured in cc's and comprising a head volume magnitude greater than 420, a head center of gravity, and an optimization characteristic. When the golf club head is at an address position over a ground plane, a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane, and a head horizontal axis extends through the head center of gravity, and is orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis extends through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the head center of gravity can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane and orthogonal to the front plane, between (a) a second intersection point located at an intersection between the front

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plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. The optimization characteristic can be defined by (a) the head volume magnitude added to (b) a ratio between the head CG depth divided by an absolute value of the head CG height. The optimization characteristic can be greater than or equal to 425.

In one example, a golf club head can comprise a head body comprising a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, where the bore can have a hosel axis. The golf club head can also comprise a strikeface at the head front portion and comprising a strikeface centerpoint, and a head center of gravity. When the golf club head is at an address position over a ground plane, a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane, and a head horizontal axis extends through the head center of gravity, and is orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis can extend through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the head center of gravity can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane and orthogonal to the front plane, between (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. An absolute value of the head CG height can be less than or equal to 2.54 mm. The head CG depth can be greater than or equal to 40.64 mm.

In one implementation, a method for providing a golf club head can comprise providing a head body comprising a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, where the bore having a hosel axis. The method can also comprise coupling a strikeface at the head front portion, and establishing an optimization characteristic of the golf club head. The strikeface comprises a strikeface centerpoint. A head volume of the golf club head can be measured in cc's and can comprise a head volume magnitude greater than 420. When the golf club head is at an address position over a ground plane, a head vertical axis can extend through the head center of gravity and can be orthogonal to the ground plane. A head horizontal axis can extend through the head center of gravity, and can be orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis can extend through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the head center of gravity can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the

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ground plane and orthogonal to the front plane, between (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. The optimization characteristic can be established by (a) the head volume magnitude added to (b) a ratio between the head CG depth divided by an absolute value of the head CG height, where the optimization characteristic can be greater than or equal to 425.

In one example, a golf club head can comprise a head body, a face portion, and a head center of gravity, and at least one of a first performance characteristic or a second performance characteristic. The head body can comprise a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, where the bore can have a hosel axis. The face portion can be at the head front portion and can comprise a strikeface centerpoint, a strikeface perimeter, and a face height bounded by the strikeface perimeter. When the golf club head is at an address position over a ground plane, a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane, and a head horizontal axis extends through the head center of gravity, and is orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis can extend through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the head center of gravity can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane and orthogonal to the front plane, between (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. The face height can be approximately 33 mm to approximately 71 mm, measured parallel to the loft plane. The first performance characteristic can comprise the head CG height being less than or equal to approximately 5.08 mm. The second performance characteristic can comprise a CG performance ratio of less than or equal to 0.56, as defined by (a) 76.2 mm minus the face height, divided by (b) the head CG depth.

In one example, a golf club head can comprise a head body, a face portion, and a head center of gravity. The head body can comprise a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, the bore having a hosel axis. The face portion can be coupled to the head front portion and can comprise a strikeface having a strikeface centerpoint, a strikeface perimeter, and a face height. When the golf club head is at an address position over a ground plane, a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane, and a head horizontal axis extends through the head center of gravity, and is orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can

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extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis can extend through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the head center of gravity can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane and orthogonal to the front plane, between (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. The face height can be approximately 33 mm to approximately 71 mm, as delimited by the strikeface perimeter and measured parallel to the loft plane. A CG performance ratio between (a) 76.2 mm minus the face height, and (b) the head CG depth, is less than or equal to 0.56. The head body can comprise a driver-type body. A head volume of the golf club head can be approximately 420 cc to approximately 470 cc. A head weight of the golf club head can be approximately 185 grams to approximately 225 grams. The head CG height can be approximately 0 mm to approximately 3.18 mm. The head CG depth can be approximately 25 mm to approximately 102 mm. The head body can comprise a weight structure located towards the sole portion and the rear portion of the head body.

In one implementation, a method for providing a golf club head can comprise providing a head body having a head front portion, a head rear portion, a head heel portion, a head toe portion, a head sole portion, a head top portion, and a hosel structure having a bore for receiving a golf club shaft, the bore having a hosel axis. The method can also comprise coupling a face portion to the head front portion, the face portion comprising a strikeface having a strikeface centerpoint, a strikeface perimeter, and a face height bounded by the strikeface perimeter. The method can further comprise establishing at least one of: a first performance characteristic of the golf club head, or a second performance characteristic of the golf club head. When the golf club head is at an address position over a ground plane, a head vertical axis extends through a head center of gravity of the golf club head and is orthogonal to the ground plane, and a head horizontal axis extends through the head center of gravity, and is orthogonal to the head vertical axis. A loft plane of the golf club head can be tangent to the strikeface centerpoint. A front plane of the golf club head can extend through the strikeface centerpoint and parallel to the hosel axis. A head depth plane can extend through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane. A CG height axis can extend through the head center of gravity and can intersect the head depth plane perpendicularly at a first intersection point. A head CG height of the golf club head can be measured, along the CG height axis, between the head center of gravity and the first intersection point. A head CG depth of the head center of gravity can be measured, parallel to the ground plane and orthogonal to the front plane, between: (a) a second intersection point located at an intersection between the front plane and the ground plane, and (b) a third intersection point located at an intersection between the head vertical axis and the ground plane. The face height can be approximately 33 mm to approximately 71 mm, measured parallel to the loft plane. The first performance characteristic can comprise the head CG height being less than or equal to approximately 5.08 mm. The second performance characteristic can com-

prise a CG performance ratio of less than or equal to 0.56, as defined by (a) 76.2 mm minus the face height, divided by (b) the head CG depth.

Other examples and embodiments are further disclosed herein. Such examples and embodiments may be found in the figures, in the claims, and/or in the present description.

Turning to the drawings, FIG. 1 illustrates a front view of golf club head 1000, comprising head body 1100 and face portion 1200. Face portion 1200 includes strikeface 1210. FIG. 2 illustrates a side cross-sectional view of golf club head 1000 along line II-II of FIG. 1. FIG. 3 illustrates a bottom view of golf club head 1000. FIGS. 1-3 present golf club head 1000 at an address position relative to ground plane 1010, where hosel axis 1710 is at a 60-degree angle with ground plane 1010 with respect to a front view of golf club head 1000 (FIG. 1), and where hosel axis 1710 is substantially orthogonal to ground plane 1010 with respect to a side view of golf club head 1000 (FIG. 2).

In the present embodiment, head body 1100 and face portion 1200 comprise separate pieces of material coupled together, for example, via a welding process. In other examples, however, face portion 1200 may comprise a single piece of material with one or more portions of head body 1100, such as head front portion 1110, head top portion 1120, head sole portion 1130, head heel portion 1140, head toe portion 1150, and/or head rear portion 2160. Head forward surface 1160 of golf club head 1000 comprises strikeface 1210, face portion 1200, and at least part of head front portion 1110. In some embodiments, head forward surface 1160 also can include at least part of head sole portion 1130. In the same or different embodiments, head front portion 1110 can include strikeface 1210 and/or face portion 1200.

Face portion 1200 comprises strikeface 1210 having strikeface centerpoint 1211, strikeface perimeter 1212, and face height 1213. Strikeface centerpoint 1211 is located at a geometric centerpoint of strikeface perimeter 1212 in the present example, and at a midpoint of face height 1213. In the same or other examples, strikeface centerpoint 1211 also can be centered with respect to engineered impact zone 1250, which can be defined by a region of grooves 1259 of strikeface 1210. As another approach, strikeface centerpoint 1211 can be located in accordance with the definition of a golf governing body such as the United States Golf Association (USGA). For example, strikeface centerpoint 1211 can be determined in accordance with Section 6.1 of the USGA's Procedure for Measuring the Flexibility of a Golf Clubhead (USGA-TPX3004, Rev. 1.0.0, May 1, 2008) (available at <http://www.usga.org/equipment/testing/protocols/Procedure-For-Measuring-The-Flexibility-Of-A-Golf-Club-Head/>) (the "Flexibility Procedure").

Golf club head 1000 comprises loft plane 2270 (FIG. 2), which is at least tangent to strikeface centerpoint 1211 at strikeface 1210. Face height 1213 can be measured parallel to loft plane 2270 between strikeface top end 1215 and strikeface bottom end 1216 of strikeface perimeter 1212, and can be of approximately 33 millimeters (mm) to approximately 71 mm in the present or other examples.

Strikeface perimeter 1212, comprising strikeface top end 1215 and strikeface bottom end 1216 defining face height 1213, need not bound an entirety of face portion 1200. For example, as seen in FIG. 1, strikeface 1210 is bounded by strikeface perimeter 1212 and is only part of face portion 1200. In some examples, strikeface 1210 can comprise a roll radius and/or a bulge radius, and strikeface perimeter 1212 can be defined along a transition boundary where a contour of face portion 1200 departs from the roll radius and/or the

bulge radius of strikeface 1210. For example, FIG. 2 contains a zoom view of part of the top transition boundary of golf club head 1000, highlighting vertical roll radius 2170 extending along strikeface 1210, and showing how strikeface top end 1215 is located at the top transition boundary where head forward surface 1160 departs from vertical roll radius 2170. FIG. 2 also contains a zoom view of part of the bottom transition boundary of golf club head 1000, highlighting vertical roll radius 2170 extending vertically along strikeface 1210, and showing how strikeface bottom end 1216 is located at the bottom transition boundary where head forward surface 1160 departs from vertical roll radius 2170.

In the same or other embodiments, strikeface perimeter 1212 can be defined with respect to the edge of a strikeplate comprising the strikeface. For instance, face portion 1200 comprises strikeplate 1220, where strikeface 1210 forms an exterior surface of faceplate 1220, and where strikeplate 1220 is joined to head front portion 1110 along strikeplate edge 1221. In the present example, strikeplate edge 1221 defines at least part of strikeface perimeter 1212, including the top and bottom sections of strikeface perimeter 1212 where strikeface top end 1215 and strikeface bottom end 1216 are respectively located to define face height 1213, but there can be other examples where the strikeplate edge of the strikeplate can define a majority or all of the strikeface perimeter of the strikeface.

As shown in FIG. 2, golf club head 1000 also comprises head center of gravity (CG) 2500, head depth plane 2310, and CG height axis 2320, where head depth plane 2310 extends through strikeface centerpoint 1211 and is perpendicular to loft plane 2270, and where CG height axis 2320 extends through head center of gravity 2500 and intersects head depth plane 2310 perpendicularly at intersection point 2801.

Head center of gravity 2500 comprises CG height 2520 and CG depth 2510, which locate head center of gravity 2500 relative to golf club head 1000. In the present example, CG height 2520 can be measured along CG height axis 2320, between head center of gravity 2500 and intersection point 2801. CG depth 2510 can be measured, as seen in FIG. 2, parallel to ground plane 1010 and between intersection points 2802-2803. In the present example, intersection point 2802 is defined by the intersection between ground plane 1010 and front plane 2280, where front plane 2280 extends through strikeface centerpoint 1211, is parallel to hosel axis 1710, and is orthogonal to ground plane 1010 when golf club head 1000 is at the address position. In addition, intersection point 2803 is defined by the intersection between ground plane 1010 and head vertical axis 1610, where head vertical axis 1610 extends through head center of gravity 2500, and is orthogonal to ground plane 1010 when golf club head 1000 is at the address position. Head center of gravity 2500 can also be located relative to ground plane 1010, where head CG elevation 2530 of head center of gravity 2500 can be measured along head vertical axis 1610, between weight center 2750 and ground plane 1010.

Head body 1100 of golf club head 1000 also comprises hosel structure 1217 (FIG. 1) and hosel axis 1710 extending along a center of a bore of hosel structure 1217. In the present example, a hosel coupling mechanism of golf club head 1000 comprises hosel structure 1217 and shaft sleeve 1411, where shaft sleeve 1411 can be coupled to an end of golf shaft 1410. Shaft sleeve 1411 can couple with hosel structure 1217 in a plurality of configurations, thereby permitting golf shaft 1410 to be secured to hosel structure 1217 at a plurality of angles relative to hosel axis 1710.

There can be other examples, however, where shaft **1410** can be non-adjustably secured to hosel structure **1217**.

Golf club heads in accordance with the present disclosure can be configured to exhibit one or more optimization characteristics that optimize or balance the performance thereof. For example, one characteristic of golf club heads that the present designs strive to optimize is that of face height and/or face size. Maximizing the face height and/or face size of a golf club head can have several benefits, such as increasing the target impact area of the strikeface to yield a more forgiving club head that produces better results for golf shots that are hit off-center of the strikeface centerpoint. In addition, a strikeface of larger height and/or size can provide for better energy transfer to the golf ball upon impact therewith, and may thus increase a characteristic time or a “spring effect” of the golf club head to achieve golf shots of longer distance. In some examples, the height or size of the strikeface may be augmented to achieve a characteristic time limit set by a golf governing body, such as the characteristic time limit of 239 microseconds (μ s) set by the USGA in its Flexibility Procedure.

Indiscriminately increasing face height and/or size, however, can adversely affect performance in other areas, such as with respect to launch angle, ball spin, and/or ball speed of the golf ball upon impact with the strikeface. For example, increasing face height and/or size can decrease the CG depth between the center of gravity and the strikeface centerpoint of the golf club head to bring the center of gravity forward, thereby reducing the dynamic loft of the golf club head and thus decreasing the launch angle for the golf ball. As another example, increasing face height and/or size can raise the CG height between the center of gravity and the head depth plane to elevate the center of gravity away from the sole of the golf club head, thereby inhibiting a gear effect between the strikeface and the golf ball, thus preventing the golf club head from decreasing the amount of backspin of the golf ball created upon impact, and thus decreasing the distance the ball will travel due to the backspin.

Considering the above, the height or size of the face of the golf club head should be balanced with respect to the location of the center of gravity. With respect to golf club head **1000**, strikeface **1210** has been increased to comprise an augmented face size and/or face height **1213** to provide a larger impact area and greater energy transfer to golf ball **2900**. In particular, golf club head **1000** can be configured so that face height **1213** can be of approximately 33 mm to approximately 71 mm to provide greater impact area and energy transfer upon impact with golf ball **2900**. In some examples, an area of strikeface **1210**, including the augmented face area, can be of approximately 23.6 centimeters squared (cm^2) to approximately 45.2 cm^2 .

Notwithstanding the augmented face size and/or face height **1213** described above, golf club head **1000** still restricts CG height **2520** from increasing towards head top portion **1120** and/or from straying too far from head depth plane **2310**. For example, golf club head **1000** comprises a first optimization characteristic satisfying Relation 1 below:

$$\text{ICG height}_{2520} \leq 5.08 \text{ mm}$$

[Relation 1]

There can be examples where CG height **2520** can be of approximately 0 mm up to Relation 1's limit of 5.08 mm. CG height **2520** can also be of up to a maximum of approximately 4.45 mm, 3.81 mm, or 3.18 mm in other examples. In some implementations, the first optimization characteristic can decrease the backspin of golf ball **2900** via a gear effect between strikeface **1210** and golf ball **2900** for

better performance. Although head center of gravity **2500** is shown in FIG. 2 as being below depth plane **2310**, such that CG height **2520** extends between depth plane **2310** and head sole portion **1130**, there can be embodiments where head center of gravity **2500** can be above depth plane **2310**, such that CG height **2520** extends between depth plane **2310** and head top portion **1120**, while still satisfying Relation 1 above.

In addition, and considering the augmented face size and/or face height **1213** described above, golf club head **1000** still restricts center of gravity **2500** from moving towards strikeface **1212**, thereby preventing CG depth **2510** from unduly decreasing. For example, golf club head **1000** comprises a second optimization characteristic satisfying Relation 2 below:

$$\frac{76.2 \text{ mm} - \text{face height}_{1213}}{\text{CG depth}_{2510}} \leq 0.56$$

[Relation 2]

Accordingly, the relationship between face height **1213** and CG depth **2510** is balanced pursuant to Relation 2 to maintain the second optimization characteristic of less than or equal to 0.56, thereby limiting the amount that CG depth **2510** can decrease towards strikeface **1210**. There can be examples where CG depth **2510** can be approximately 25 mm to approximately 102 mm. In the same or other examples, CG depth **2510** can be at least approximately 39 mm. In some implementations, the second optimization characteristic can increase or optimize at least one of a dynamic loft of golf club head **1000** or a launch angle of golf ball **2900** upon impact therebetween.

In some examples, golf club head **1000** may be configured to comprise only one of the first or second optimization characteristics described above. For example, golf club head **1000** may comprise the first optimization characteristic and not the second optimization characteristic, thus satisfying Relation 1 without having to satisfy Relation 2. As another example, golf club head **1000** may comprise the second optimization characteristic and not the first optimization characteristic, thus satisfying Relation 2 without having to satisfy Relation 1. In addition, there can be embodiments where golf club head **1000** satisfies both Relations 1 and 2, and thus comprises the first and second optimization characteristics.

Golf club head **1000** can also comprise a third optimization characteristic with respect to head volume (HV) **2600** thereof. In the present example, head body **1000** of golf club head **1000** comprises a driver-type body with a head volume greater than or equal to 420 cubic centimeters (cc), and thus has a head volume magnitude greater than or equal to 420. For example, head body **1000** can comprise a head volume of 420 cc, thus having a head volume magnitude of 420. As another example, golf club head **1000** can comprise a head volume of 460 cc, thus having a head volume magnitude of 460. Golf club head **1000** can comprise a head volume up to approximately 470 cc, in some implementations, and/or a total head weight of approximately 185 grams to approximately 225 grams. In some specific examples, the total head weight can be approximately 202 grams, and/or the head volume can be of approximately 460 cc.

The third optimization characteristic can control the relationship between head volume **2600** and the location of center of gravity **2500**, and can be defined to satisfy Relation 3 below:

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$$HV + \frac{CG \text{ depth}_{2510}}{|CG \text{ height}_{2520}|} \geq 425 \quad [\text{Relation } 3]$$

In some instances, head volume **2600** can be increased to adjust, for instance, a moment of inertia (MOI) of golf club head **1000**. An unrestrained increase in head volume, however, can have detrimental effects with respect to other characteristics of the golf club head. For example, increasing head volume **2600** can cause head center of gravity **2500** to shift towards head front portion **1110**, towards head top portion **1120**, towards other undesired directions, and/or away from a desired center of gravity location or direction (s), thereby hampering the performance of golf club head **1000**. Such undesired changes in the center of gravity location can detrimentally affect one or more characteristics of the golf club head, such as launch speed, launch angle, gear effect, backspin, and or shot distance. Accordingly, the third optimization characteristic used to balance the relationship between head volume **2600** and the location of head center of gravity **2500** can be established to yield desirable and balanced attributes for golf club head **1000**. For instance, a weight distribution of golf club head **1000** can be configured to satisfy Relation 3 so that golf club head **1000** can exhibit the third optimization characteristic, thereby permitting head volume **2600** to be augmented for greater moment of inertia and greater energy transfer to golf ball **2900** upon a golf impact between strikeface **1210** and golf ball **2900**. In the same or other implementations, the weight distribution of golf club head **1000** can be configured for restricting CG depth **2510** from decreasing towards head front portion **1110** due to the augmented head volume **2600**, thus increasing at least one of a dynamic loft of strikeface **1210** or a launch angle of golf ball **2900** upon the golf impact. In addition, the weight distribution of golf club head **1000** can be configured for restricting CG height **2520** from increasing towards head top portion **1120** as a result of the augmented head volume **2600**, thereby decreasing a backspin of golf ball **2900** via a gear effect between strikeface **1210** and golf ball **2900** upon the golf impact.

Considering the above, to attain the third optimization characteristic in compliance with Relation 3, CG depth **2510** can be configured to be greater than or equal to 40.64 mm. In the same or other embodiments, the absolute value of CG height **2520** can be less than or equal to 2.54 mm. Note that CG height **2520** is characterized as an absolute value, considering that head center of gravity **2500** can be above or below head depth plane **2310** in some embodiments. Although the third optimization characteristic has a lower bound of at least 425, there can be other embodiments where the third optimization characteristic can be defined with respect to other lower bounds. For instance, the third optimization characteristic can comprise a lower bound of at least 435 or 445 in some implementations. The location of head center of gravity **2500** can also be designed or configured with respect to other features of golf club head **1000** in order to satisfy Relation 3 and/or to attain the third optimization characteristic. For instance, the location of head center of gravity **2500** can be configured such that CG depth **2510** comprises between approximately 25% to approximately 80% of head depth length **2312**, where head depth length **2312** is measured from strikeface centerpoint **1211** to an intersection of an exterior of head rear portion **2160** by head depth plane **2310**. As another example, the location of head center of gravity **2500** can be configured such that CG height **2520** comprises between approximately 0% to

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approximately 13% of CG height axis length **2322**, where CG height axis length **2322** is measured from an intersection of an exterior of head top portion **1120** by CG height axis **2320**, to an intersection of an exterior of head sole portion **1130** by CG height axis **2320**.

Golf club head **1000** also can comprise a fourth optimization characteristic with respect to a balance between hosel MOI **1711** (FIG. 1) and horizontal MOI **1811** (FIG. 1). Hosel MOI **1711** is defined about hosel axis **1710**. Horizontal MOI **1811** is defined about head horizontal axis **1810**, which extends through head center of gravity **2500**, from head heel portion **1140** to head toe portion **1150**, and parallel to ground plane **1010** when golf club head **1000** is at the address position over ground plane **1010**.

In some examples, horizontal MOI **1811** can be increased to restrict a rotation of golf club head **1000** about head horizontal axis **1810** when strikeface **1210** hits golf ball **2600** off-center towards head top portion **1120** or head sole portion **1130**, thereby increasing the forgiveness of golf club head **1000** for such high or low mis-hits. For instance, to increase horizontal MOI **1811**, weight may be added or repositioned towards head front portion **1110** and/or head rear portion **2160**. In the same or other examples, golf club head **1000** can be lengthened towards head front portion **1110** and/or head rear portion **2160**.

Such adjustments or changes to increase horizontal MOI **1811** can be made up to a point, however, before they start affecting other golf club head characteristics. For example, unrestrained adjustments to increase horizontal MOI **1811** can lead to an undue increase in hosel MOI **1711** if not properly balanced, thereby increasing the resistance of golf club head **1000** to rotate about hosel axis **1710**, and thus making it hard for a person to “turn over” the golf club during a golf swing for proper positioning or “squaring” of golf club head **1000** at impact with golf ball **2600**. An increase in hosel MOI **1711** also can restrict or reduce a gearing effect between golf ball **2600** and strikeface **1210** that would otherwise impart some corrective spin to golf ball **2600** during off-center impacts.

To reduce hosel MOI **1711**, golf club head **1000** can be designed to limit the distance between hosel axis **1710** and any additional or discretionary mass of golf club head **1000**. Such approaches to decrease hosel MOI **1711**, if not properly balanced, can be incompatible with some of the approaches described above to increase horizontal MOI **1811**. Accordingly, weight addition or redistribution for golf club head **1000** to increase horizontal MOI **1811** should be balanced with respect to maintaining or restricting an increase in hosel MOI **1711**.

In light of the above, the fourth optimization characteristic of golf club head **1000** controls the relationship between horizontal MOI **1811** and hosel MOI **1711** to satisfy to satisfy Relation 4 below:

$$(\text{Horizontal MOI}_{1811}) \geq 39\% (\text{Hosel MOI}_{1711}) \quad [\text{Relation } 4]$$

There can be examples where golf club head **1000** can be configured so that its fourth optimization characteristic can surpass the requirements of Relation 4. As an example, in some implementations, the fourth optimization characteristic of golf club head **1000** can be configured so that horizontal MOI **1811** is greater than or equal to 40% of hosel MOI **1711**, greater than or equal to 45% of hosel MOI **1711**, or greater than or equal to 50% of hosel MOI **1711**. In the present example, horizontal MOI **1811** is approximately 3740 grams-square-centimeter (g-cm²), but there can be examples where it can range between approximately 2800 g-cm² to approximately 4300 g-cm². Hosel MOI **1711** is

approximately 9370 g·cm² in the present example, but can range between approximately 7000 g·cm² and approximately 11,000 g·cm² in the same or other examples.

Golf club head 1000 also can comprise a fifth optimization characteristic with respect to a balance between hosel MOI 1711 and vertical MOI 1611 (FIG. 1). Vertical MOI 1611 is defined about head vertical axis 1610, which extends through head center of gravity 2500, from head top portion 1120 to head sole portion 1130, and orthogonal to ground plane 1010 when golf club head 1000 is at the address position. Vertical MOI 1611 is approximately 5300 g·cm² in the present example, but can range between approximately 4700 g·cm² and approximately 6000 g·cm² in the same or other examples.

In some examples, vertical MOI 1611 can be increased to restrict a rotation of golf club head 1000 about head vertical axis 1610 when strikeface 1210 hits golf ball 2600 off-center towards head heel portion 1140 or towards head toe portion 1150, thereby increasing the forgiveness of golf club head 1000 for such heel-side or toe-side mis-hits. For instance, to increase vertical MOI 1611, weight can be added or repositioned towards head heel portion 1140 and/or head toe portion 1150. In the same or other examples, the golf club head can be lengthened towards head heel portion 1140 and/or head toe portion 1150.

Such adjustments or changes to increase vertical MOI 1611 can be made up to a point, however, before they start affecting other golf club head characteristics. For example, unrestrained adjustments to increase vertical MOI 1611 can lead to undue increase in hosel MOI 1711 if not properly balanced, thereby increasing the resistance of golf club head 1000 to rotate about hosel axis 1710 as described above. In addition, some approaches to decrease hosel MOI 1711, if not properly balanced, can be incompatible with some of the approaches described above to increase vertical MOI 1611. Accordingly, weight addition or redistribution for golf club head 1000 to increase vertical MOI 1611 should be balanced with respect to maintaining or restricting an increase in hosel MOI 1711.

In light of the above, the fifth optimization characteristic of golf club head 1000 controls the relationship between vertical MOI 1611 and hosel MOI 1711 to satisfy Relation 5 below:

$$(\text{Vertical MOI}_{1611}) \geq 59\% (\text{Hosel MOI}_{1711}) \quad [\text{Relation 5}]$$

There can be examples where golf club head 1000 can be configured so that its fifth optimization characteristic can surpass the requirements of Relation 5. As an example, in some implementations, the fifth optimization characteristic of golf club head 1000 can be configured so that vertical MOI 1611 is greater than or equal to 60% of hosel MOI 1711, greater than or equal to 65% of hosel MOI 1711, or greater than or equal to 70% of hosel MOI 1711. In some examples, golf club head 1000 can be configured so that its fourth optimization characteristic satisfies Relation 4, while its fifth optimization characteristic also satisfies Relation 5.

In some implementations, golf club head 1000 can be configured to exhibit the first, second, third, fourth, and/or fifth optimization characteristics described above by adjusting a distribution of mass or a relationship between different elements of golf club head 1000. To such ends, golf club head 1000 can comprise weight structure 2700, located towards head sole portion 1130 and head rear portion 2160, as seen in FIGS. 2-3. In some configurations, weight structure 2700 can be designed and/or located to satisfy the constraints imposed by Relation(s) 1, 2, 3, 4, and/or 5, thereby balancing the face height or size of strikeface 1210,

head volume 2600, the location of center of gravity 2500, and/or the different moments of inertia of golf club head 1000.

As can be seen in FIG. 3, weight structure 2700 can be located relative to clock grid 3500, which can be aligned with respect to strikeface 1210. For example, clock grid 3500 comprises 12 o'clock ray 3512, which is aligned with strikeface centerpoint 1211 in the present embodiment. 12 o'clock ray 3512 is orthogonal to front intersection line 3271, which is defined by the intersection of loft plane 2270 (FIGS. 2-3) and ground plane 1010 (FIGS. 1-2). Clock grid 3500 can be centered along 12 o'clock ray 3512, at a midpoint between a front end of front portion 1110 and a rear end of rear portion 2160. In the same or other examples, clock grid centerpoint 3515 can be centered proximate to a geometric centerpoint of golf club head 1000. Clock grid 3500 also comprises 3 o'clock ray 3503 extending towards head heel portion 1140, and 9 o'clock ray 3509 extending towards head toe portion 1150.

Weight perimeter 2705 of weight structure 2700 is located in the present embodiment towards head rear portion 2160, at least partially bounded between 4 o'clock ray 3504 and 8 o'clock ray 3508 of clock grid 3500, while weight center 2750 is located between 5 o'clock ray 3505 and 7 o'clock ray 3507. In examples such as the present one, weight perimeter 2705 is fully bounded between 4 o'clock ray 3504 and 8 o'clock ray 3508. Although weight perimeter 2705 is defined external to golf club head 1000 in the present example, there can be other examples where weight perimeter may extend into an interior of, or be defined within, golf club head 1000. In some examples, the location of weight 2700 can be established with respect to a broader area. For instance, in such examples, weight perimeter 2705 of weight structure 2700 can be located towards head rear portion 2160, at least partially bounded between 4 o'clock ray 3504 and 9 o'clock ray 3509 of clock grid 3500, while weight center 2750 can be located between 5 o'clock ray 3505 and 8 o'clock ray 3508.

In the same or other embodiments, weight structure 2700 can extend or be shifted towards heel portion 1140. For instance, weight perimeter 2705 and/or weight center 2750 can be shifted towards 4 o'clock ray 3504 than towards 9 o'clock ray 3509. Biasing weight structure 2700 towards head heel end 1140 can permit a decrease in hosel MOI 1711 about hosel axis 1710 by limiting the distance between hosel axis 1710 and weight structure 2700, thereby allowing easier turning of golf club head 1000 about hosel axis 1710 during a swing.

In some examples, weight structure 2700 can comprise a mass of approximately 2 grams to approximately 50 grams, and/or a volume of approximately 1 cc to approximately 30 cc. In the present example, weight structure 2700 protrudes from the external contour of head sole portion 1130, and is thus at least partially external to allow for greater adjustment of head center of gravity 2500.

Weight structure 2700 can comprise removable weight 2790 in the same or other examples, where removable weight 2790 can comprise a mass of approximately 0.5 grams to approximately 30 grams, and can be replaced with one or more other similar weights to adjust the location of head center of gravity 2500 if needed to satisfy Relation(s) 1, 2, 3, 4, and/or 5. In the same or other examples, weight center 2750 can comprise at least one of a center of gravity of weight structure 2700, a center of gravity of removable weight 2790, a geometric center of weight structure 2700, and/or a geometric center of removable weight 2790.

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Weight center **2750** can be located with respect to ground plane **1010** and weight center elevation axis **2340**, which extends between weight center **2750** and ground plane **1010**. Weight center elevation axis **2340** is orthogonal to ground plane **1010** when golf club head **1000** is at the address position. Weight center elevation **2730** for weight center **2750** can thus be measured along weight center elevation axis **2340**, between weight center **2750** and ground plane **1010**. In addition, weight center depth **2710** for weight center **2750** can be measured, parallel to ground plane **1010**, between intersection points **2802** and **2804**. In the present example, intersection point **2804** is defined by the intersection between ground plane **1010** and weight center elevation axis **2340** when golf club head **1000** is at the address position. Weight center **2750** can be located in the same or other embodiments such that weight distance **2751** (FIG. 2), which separates head center of gravity **2500** from weight center **2750**, can be approximately 25 mm to approximately 102 mm.

There can also be embodiments where face portion **1200** can comprise a reduced thickness, which may be reinforced as needed with one or more reinforcing structures at the backside of strikeface **1210** and/or at the junction between face portion **1200** and head front portion **1110**. Other mass redistribution mechanisms can be employed as well if desired to satisfy Relation(s) 1, 2, 3, 4, and/or 5.

In some implementations, a relationship or ratio between head center of gravity **2500** and weight center **2750** can be configured to permit one or more of Relation(s) 1, 2, 3, 4, or 5 to be satisfied. For example, an elevation ratio, defined by the ratio of weight center elevation **2730** over head CG elevation **2530**, can be greater than 0.44 to help maintain head center of gravity **2500** closer to head sole portion **1130**. As another example, a depth ratio, defined by the ratio of weight center depth **2710** over head CG depth **2510**, can be less than 2.54 to preventing CG depth **2510** from unduly decreasing towards head front portion **1110**. There can be some implementations where head CG elevation **2530** can be less than approximately 28.5 mm, where weight center elevation **2730** can be less than approximately 12.5 mm, and/or where weight center depth **2710** can be greater than approximately 99.7 mm.

FIG. 4 illustrates a flowchart for method **4000**, which can be used to provide, form, and/or manufacture a golf club head in accordance with the present disclosure. In some examples, the golf club head can be similar to golf club head **1000** (FIGS. 1-3) presented above.

Method **4000** comprises block **4100** for providing a head body of a golf club head comprising a head front portion. In some examples, the head body can be similar to head body **1100** (FIGS. 1-3), and the head front portion can be similar to head front portion **1110** (FIGS. 1-3).

Block **4200** of method **4000** comprises coupling a face portion to the head front portion, the head front portion comprising a strikeface with an augmented face size. In some examples, the face portion can be similar to face portion **1200** (FIGS. 1-2), with strikeface **1210** having the augmented face size described above with respect thereto. For example, the augmented face size of the strikeface may permit its face height to be of up to approximately 71 mm in some examples.

Method **4000** can comprise block **4300** for configuring the golf club head to comprise a first optimization characteristic, where a CG height between a center of gravity of the golf club head and a head depth plane of the golf club head can be approximately 0 mm to approximately 5.08 mm or 0.200 inches. In some examples, the first optimization character-

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istic can be similar to that described above with respect to Relation 1 for balancing golf club head face height or size with respect to center of gravity height. In some examples, the CG height may be similar to CG height **2520** (FIG. 2); the center of gravity may be similar to head center of gravity **2500** (FIG. 2); and the head depth plane can be similar to head depth plane **2310** (FIG. 2).

There can be implementations where method **4000** can comprise block **4400** for configuring the golf club head to comprise a second optimization characteristic, where a ratio between (a) 76.2 mm (or approximately 3.0 inches) minus the face height and (b) a CG depth between the strikeface centerpoint and the center of gravity, is less than 0.56. In some examples, the second optimization characteristic can be similar to that described above with respect to Relation 2 for balancing golf club head face height or size with respect to center of gravity depth. For example, the face height can be similar to face height **1213**, and the CG depth can be similar to CG depth **2510**.

In some examples, method **4000** can comprise block **4500** for configuring the golf club head to comprise a third optimization characteristic where a head volume magnitude, added to a ratio between the CG depth and the CG height, is greater than or equal to 425. In some implementations, the third optimization characteristic can be similar to that described above with respect to Relation 3 for balancing head volume relative to center of gravity location. For example, the head volume magnitude can be similar to the magnitude of head volume **2600** (FIG. 2), the CG depth can be similar to CG depth **2510**, and the CG height can be similar to CG height **2520**.

Method **4000** can comprise block **4600** in some embodiments for configuring the golf club head to comprise a fourth optimization characteristic, where the golf club head's horizontal moment of inertia is greater than or equal to 39% of its hosel moment of inertia. In some implementations, the fourth optimization characteristic can be similar to that described above with respect to Relation 4 for balancing horizontal MOI **1811** with respect to hosel MOI **1711** (FIG. 1). In the same or other examples, the magnitude of the horizontal moment of inertia can be similar to that described above with respect to horizontal MOI **1811**. In addition, the magnitude of the hosel moment of inertia can be similar to that described above with respect to hosel MOI **1711**. There can also be examples where the horizontal moment of inertia and/or the hosel moment of inertia can be balanced with respect to other features, such as with respect to a vertical moment of inertia of the golf club head.

Block **4700** of method **4000** can be carried out in some implementations for configuring the golf club head to comprise a fifth optimization characteristic, where the golf club head's vertical moment of inertia is greater than or equal to 59% of its hosel moment of inertia. In some implementations, the fifth optimization characteristic can be similar to that described above with respect to Relation 5 for balancing vertical MOI **1611** with respect to hosel MOI **1711** (FIG. 1). In the same or other examples, the magnitude of the vertical moment of inertia can be similar to that described above with respect to vertical MOI **1611**. In addition, the magnitude of the hosel moment of inertia can be similar to that described above with respect to hosel MOI **1711**. There can also be examples where the vertical moment of inertia and/or the hosel moment of inertia can be balanced with respect to other features, such as with respect to the horizontal moment of inertia of block **4500**.

In the present example, method **4000** also comprises block **4800** for providing a mass redistribution mechanism

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to adjust the center of gravity of the golf club head. In some examples, the mass redistribution mechanism can be configured to permit the golf club head to achieve the requirements of block 4300, block 4400, block 4500, block 4600, and/or block 4700 of method 4000. The mass redistribution mechanism can comprise a weight structure, such as weight structure 2700 (FIGS. 2-3), which can adjust the location of the center of gravity towards the sole and/or the rear portion of the golf club head if desired. In the same or other embodiments, the mass redistribution mechanism can comprise a reduced thickness of the face portion of the golf club head, which may be reinforced if needed with one or more reinforcing structures, such as at the backside of the strikeface, and/or at a junction between the face portion and the head body of the golf club head.

In some examples, one or more of the different blocks of method 4000 can be combined into a single block or performed simultaneously, and/or the sequence of such blocks can be changed. For example, blocks 4100 and 4200 may be combined in some embodiments, such as where the face portion and at least one portion of the head body comprise a single piece of material. Block 4800 may be combined with one or more of blocks 4100, 4300, 4400, 4500, 4600, and/or 4700 in the same or other examples, and may be achieved simultaneously by adjusting the center of gravity, the face height, the face size, the head volume, and/or one or more moments of inertia of the golf club head, such as via the mass redistribution mechanism of block 4800. In the same or other examples, some of the blocks of method 4000 can be subdivided into several sub-blocks. For example, block 4100 can be subdivided into several sub-blocks for providing different portions of the head body of the golf club head. There can also be examples where method 4000 can comprise further or different blocks. As an example, method 4000 may comprise another block for providing or coupling a golf club shaft to the head body of block 4100. In addition, there may be examples where method 4100 can comprise only part of the blocks described above. For example, one or more of blocks 4300, 4400, 4500, 4600, and/or 4700 may be optional in some implementations, and/or block 4800 may be skipped if not needed to achieve the requirements of block 4300, block 4400, block 4500, block 4600, and/or block 4700. Other variations can be implemented for method 4000 without departing from the scope of the present disclosure.

Although the golf club heads with optimized characteristics and related methods herein have been described with reference to specific embodiments, various changes may be made without departing from the spirit or scope of the present disclosure. For instance, while the above examples may be described in connection with a driver-type golf club, the apparatus, methods, and articles of manufacture described herein may be applicable to other types of golf club such as a fairway wood-type golf club, a hybrid-type golf club, an iron-type golf club, a wedge-type golf club, or a putter-type golf club. Alternatively, the apparatus, methods, and articles of manufacture described herein may be applicable other type of sports equipment such as a hockey stick, a tennis racket, a fishing pole, a ski pole, etc.

Additional examples of such changes and others have been given in the foregoing description. Other permutations of the different embodiments having one or more of the features of the various figures are likewise contemplated. Accordingly, the specification, claims, and drawings herein are intended to be illustrative of the scope of the disclosure and is not intended to be limiting. It is intended that the

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scope of this application shall be limited only to the extent required by the appended claims.

The golf club heads with optimized characteristics and related methods discussed herein may be implemented in a variety of embodiments, and the foregoing discussion of certain of these embodiments does not necessarily represent a complete description of all possible embodiments. Rather, the detailed description of the drawings, and the drawings themselves, disclose at least one preferred embodiment, and may disclose alternative embodiments.

Replacement of one or more claimed elements constitutes reconstruction and not repair. Additionally, benefits, other advantages, and solutions to problems have been described with regard to specific embodiments. The benefits, advantages, solutions to problems, and any element or elements that may cause any benefit, advantage, or solution to occur or become more pronounced, however, are not to be construed as critical, required, or essential features or elements of any or all of the claims, unless such benefits, advantages, solutions, or elements are expressly stated in such claims.

As the rules to golf may change from time to time (e.g., new regulations may be adopted or old rules may be eliminated or modified by golf standard organizations and/or governing bodies such as the United States Golf Association (USGA), the Royal and Ancient Golf Club of St. Andrews (R&A), etc.), golf equipment related to the apparatus, methods, and articles of manufacture described herein may be conforming or non-conforming to the rules of golf at any particular time. Accordingly, golf equipment related to the apparatus, methods, and articles of manufacture described herein may be advertised, offered for sale, and/or sold as conforming or non-conforming golf equipment. The apparatus, methods, and articles of manufacture described herein are not limited in this regard.

Moreover, embodiments and limitations disclosed herein are not dedicated to the public under the doctrine of dedication if the embodiments and/or limitations: (1) are not expressly claimed in the claims; and (2) are or are potentially equivalents of express elements and/or limitations in the claims under the doctrine of equivalents.

The invention claimed is:

1. A golf club head comprising:

a head body comprising:

a head interior bounded by:

a head front portion;

a head rear portion;

a head heel portion;

a head toe portion;

a head sole portion; and

a head top portion;

a hosel structure having a bore for receiving a golf club shaft, the bore having a hosel axis;

a head volume magnitude; and

an external weight structure located towards the head sole portion and the head rear portion of the head body, the external weight structure protruding at least partially from an external contour of the head sole portion of the head body and having a weight center;

a face portion at the head front portion and comprising:

a strikeface centerpoint;

a strikeface perimeter; and

a face height bounded by the strikeface perimeter, the strikeface centerpoint positioned at a midpoint of the face height; and

a head center of gravity;

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wherein:

- when the golf club head is at an address position over a ground plane;
- a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane;
- a head horizontal axis extends through the head center of gravity, from the head heel portion to the head toe portion and is orthogonal to the head vertical axis;
- a loft plane of the golf club head is tangent to the strikeface centerpoint;
- a front plane of the golf club head extends through the strikeface centerpoint, parallel to the hosel axis, and orthogonal to the ground plane;
- a head depth plane extends through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane;
- a CG height axis extends through the head center of gravity and intersects the head depth plane perpendicularly at a first intersection point;
- a CG height axis length is measured from an intersection of the head top portion exterior with the CG height axis to an intersection of the head sole portion exterior with the CG height axis;
- a head CG height of the head center of gravity is measured, along the CG height axis, between the head center of gravity and the first intersection point;
- a head CG depth of the head center of gravity is measured, parallel to the ground plane and orthogonal to the front plane, between:
 - a second intersection point located at an intersection between the front plane and the ground plane; and
 - a third intersection point located at an intersection between the head vertical axis and the ground plane;
- a hosel moment of inertia is measured about the hosel axis;
- a horizontal moment of inertia is measured about the head horizontal axis;
- a vertical moment of inertia measured about the head vertical axis;
- the head CG depth is approximately 35 mm to approximately 102 mm;
- the face height is approximately 40 mm to approximately 50 mm, measured parallel to the loft plane; and
- the golf club head comprises at least three performance characteristics selected from the group consisting of: a first performance characteristic, a second performance characteristic, a third performance characteristic, a fourth performance characteristic, and a fifth performance characteristic; wherein:
 - the first performance characteristic comprises:
 - the head CG height being less than or equal to approximately 5.08 mm;
 - the second performance characteristic comprises:
 - a CG performance ratio less than or equal to 0.56, as defined by (a) 76.2 mm minus the face height, divided by (b) the head CG Depth;
 - the third performance characteristic comprises:
 - (a) the head volume magnitude added to (b) a ratio between the head CG depth divided by an absolute value of the head CG height; wherein: the third performance characteristic is greater than or equal to 425 cc;

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the fourth performance characteristic comprises:

- the horizontal moment of inertia being greater than or equal to 39% of the hosel moment of inertia; and

the fifth performance characteristic comprises:

- the vertical moment of inertia being greater than or equal to 59% of the hosel moment of inertia;

a weight center elevation axis extends between the weight center and the ground plane, being orthogonal to the ground plane when the golf club head is at the address position;

a head CG elevation is measured, along the head vertical axis, between the head center of gravity and the ground plane;

a weight center elevation is measured, along the weight center elevation axis, between the head center of gravity and the ground plane;

the weight center elevation is less than approximately 12.5 mm;

a weight center depth of the weight center is measured, parallel to the ground plane, between:

- the second intersection point at the intersection between the front plane and the ground plane; and
- a fourth intersection point at an intersection between the weight center elevation axis and the ground plane;

a depth ratio, defined by the weight center depth over the head CG depth, is less than 2.54; and

an elevation ratio, defined by the weight center elevation over the head CG elevation, is greater than 0.44.

2. The golf club head of claim 1, wherein:

the horizontal moment of inertia is approximately 2800 g*cm² to approximately 4300 g*cm².

3. The golf club head of claim 1, wherein:

the weight center depth is greater than approximately 99.7 mm.

4. The golf club head of claim 1, wherein:

the head CG depth is approximately 44 mm to approximately 102 mm.

5. The golf club head of claim 1, wherein:

the head volume magnitude greater than or equal to 420 cc.

6. The golf club head of claim 1, wherein the head body comprises:

a total head weight of approximately 185 grams to approximately 225 grams.

7. The golf club head of claim 1, wherein:

the head CG height is approximately 0 mm to 3.18 mm.

8. The golf club head of claim 1, wherein:

the hosel moment of inertia is between approximately 7000 g*cm² and approximately 11000 g*cm².

9. The golf club head of claim 1 wherein:

the head center of gravity is located between the head depth plane and the head sole portion.

10. The golf club head of claim 1, wherein:

the face height is approximately 40 mm to approximately 50 mm, measured parallel to the loft plane.

11. A golf club head comprising:

a head body comprising:

- a head interior bounded by:
 - a head front portion;
 - a head rear portion;
 - a head heel portion;
 - a head toe portion;
 - a head sole portion; and
 - a head top portion;

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a hosel structure having a bore for receiving a golf club shaft, the bore having a hosel axis;
 a head volume magnitude; and
 an external weight structure located towards the head sole portion and the head rear portion of the head body, the external weight structure protruding at least partially from an external contour of the head sole portion of the head body and having a weight center;

a face portion at the head front portion and comprising:
 a strikeface centerpoint;
 a strikeface perimeter; and
 a face height bounded by the strikeface perimeter, the strikeface centerpoint positioned at a midpoint of the face height; and
 a head center of gravity;

wherein:
 when the golf club head is at an address position over a ground plane:
 a head vertical axis extends through the head center of gravity and is orthogonal to the ground plane;
 a head horizontal axis extends through the head center of gravity, from the head heel portion to the head toe portion and is orthogonal to the head vertical axis;
 a loft plane of the golf club head is tangent to the strikeface centerpoint;
 a front plane of the golf club head extends through the strikeface centerpoint, parallel to the hosel axis, and orthogonal to the ground plane;
 a head depth plane extends through the strikeface centerpoint, parallel to the head horizontal axis and perpendicular to the loft plane;
 a CG height axis extends through the head center of gravity and intersects the head depth plane perpendicularly at a first intersection point;
 a CG height axis length is measured from an intersection of the head top portion exterior with the CG height axis to an intersection of the head sole portion exterior with the CG height axis;
 a head CG height of the head center of gravity is measured, along the CG height axis, between the head center of gravity and the first intersection point;
 a head CG depth of the head center of gravity is measured, parallel to the ground plane and orthogonal to the front plane, between:
 a second intersection point located at an intersection between the front plane and the ground plane; and
 a third intersection point located at an intersection between the head vertical axis and the ground plane;
 a hosel moment of inertia is measured about the hosel axis;
 a horizontal moment of inertia is measured about the head horizontal axis;
 a vertical moment of inertia measured about the head vertical axis;
 the head CG depth is approximately 35 mm to approximately 102 mm;
 the face height is approximately 40 mm to approximately 50 mm, measured parallel to the loft plane; and
 the golf club head comprises at least three performance characteristics selected from the group consisting of: a first performance characteristic, a second performance characteristic, a third performance characteristic, a fourth performance characteristic, and a fifth performance characteristic; wherein:

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the first performance characteristic comprises:
 the head CG height being less than or equal to approximately 5.08 mm;

the second performance characteristic comprises:
 a CG performance ratio less than or equal to 0.56, as defined by (a) 76.2 mm minus the face height, divided by (b) the head CG Depth;

the third performance characteristic comprises:
 (a) the head volume magnitude added to (b) a ratio between the head CG depth divided by an absolute value of the head CG height;

wherein:
 the third performance characteristic is greater than or equal to 425 cc;

the fourth performance characteristic comprises:
 the horizontal moment of inertia being greater than or equal to 39% of the hosel moment of inertia; and

the fifth performance characteristic comprises:
 the vertical moment of inertia being greater than or equal to 59% of the hosel moment of inertia;

a weight center elevation axis extends between the weight center and the ground plane, being orthogonal to the ground plane when the golf club head is at the address position;

a head CG elevation is measured, along the head vertical axis, between the head center of gravity and the ground plane;

a weight center elevation is measured, along the weight center elevation axis, between the head center of gravity and the ground plane;

a weight center depth of the weight center is measured, parallel to the ground plane, between:
 the second intersection point at the intersection between the front plane and the ground plane;
 a fourth intersection point at an intersection between the weight center elevation axis and the ground plane; and
 wherein the weight center depth is greater than approximately 99.7 mm;

a depth ratio, defined by the weight center depth over the head CG depth, is less than 2.54; and
 an elevation ratio, defined by the weight center elevation over the head CG elevation, is greater than 0.44.

12. The golf club head of claim 11 wherein:
 The head CG elevation is less than 28.5 mm.

13. The golf club head of claim 11 wherein:
 the hosel moment of inertia is between 7000 g*cm² and 11,000 g*cm².

14. The golf club head of claim 11 wherein:
 the external weight structure comprises a mass of approximately 2 grams to 50 grams; and
 the external weight structure comprises a volume of approximately 1 cc to 30 cc.

15. The golf club head of claim 11, wherein:
 a clock grid comprises at least:
 a 12 o'clock ray;
 a 3 o'clock ray;
 a 4 o'clock ray;
 a 5 o'clock ray;
 an 8 o'clock ray; and
 a 9 o'clock ray;

when the golf club head is at the address position over the ground plane, from a bottom view of the golf club head, the 12 o'clock ray is aligned with the strikeface centerpoint and orthogonal to a front intersection line between the loft plane and the ground plane;

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the clock grid is centered along the 12 o'clock ray, at a
midpoint between a front end of the head front portion
and a rear end of the head rear portion;
the 3 o'clock ray extends towards the head heel portion;
the 9 o'clock ray extends towards the head toe portion; 5
a perimeter of the external weight structure is at least
partially bounded by the 4 o'clock ray and the 9 o'clock
ray; and
a center of gravity of the external weight structure is
located between the 5 o'clock ray and the 8 o'clock ray. 10

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